

22. (1.5 points)

An insurance company has decided to manage the underwriting cycle by reducing market share when pricing is soft and expanding market share when pricing is hard.

a. (1 point)

Outline and justify an asset management strategy that could reduce the company's earnings volatility.

b. (0.5 point)

Discuss a risk that would increase if this strategy were implemented.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 22	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: C1: Demonstrate how insurance and financial risk can be analyzed quantitatively; C2: Describe the rationale for, methods for, and effect of managing insurance and financial risks; and C8: Describe approaches to modeling the underwriting cycle.
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> Invest more in high yielding assets such as equity and high yield corporate bonds during soft market, and invest in more conservative assets such as treasury during hard market. Because during soft market, company is taking on less insurance risk by reducing market share, so it makes to take on more asset risk, and the extra investment income would help offset the reduction in UW income. During hard market it's the other way around. <i>Sample Answer 2</i> Shift assets more to equities when market is soft and move to bonds when market is hard. Equities typically have higher returns than bonds, so they should help make up for the decrease in UW profit in soft market. Conversely, higher UW profit in hard market will be offset by lower investment returns from bond-heavy asset portfolio. Should smooth out annual earnings. <i>Sample Answer 3</i> During soft market, invest more in taxable bonds with higher returns. During hard market, invest more in tax exempt bonds with lower return. Justification: 1. During soft market, company suffers UW loss. The higher investment income can help offset the underwriting loss, improving performance. 2. During hard market, company with decent UW profit can use tax exempt bonds to pay less tax on the investment income from tax exempt bonds.	
Part b: 0.5 point (each corresponds to the sample answer for part a. above)	
<i>Sample Answer 1</i> Asset risk would increase during soft markets – equities are riskier than bonds; there's a risk that market prices would decline after you invest more heavily in equities. <i>Sample Answer 2</i> Investment risk/asset risk. This is risk that company may see a drop in asset value if there's a market downturn, because now the company is investing more in higher risk asset. <i>Sample Answer 3</i> Taxable bonds with higher returns might have a longer duration, which would increase the interest rate risk.	
EXAMINER'S REPORT	
○ This item was challenging due to its open-ended wording and the requirement to synthesize	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

understanding of both the underwriting cycle and asset management. ○ The item's requirement to "outline and justify" a strategy requires a high level of understanding in order to construct and justify a specific practical strategy for this situation that goes beyond a simple description.
Part a
○ Candidates needed to know that ▪ UW profits go down during soft market and increase during hard market. ▪ The company can change its investments to provide higher returns during soft market and to accept lower risk investment returns to offset the higher underwriting profit and reduce risk during the hard market. ▪ The candidate was expected to provide an asset management/financial investment strategy that could stabilize earnings across the cycle; given that this company is managing the cycle by decreasing market share during soft market and increasing market share during the hard market. ▪ What kind of assets would provide the require returns at different times in the cycle ▪ Relatively few candidates obtained full credit. ○ Common errors: ▪ Many candidates did not outline a specific asset management strategy to address this situation, and instead described general approaches to asset-liability management, portfolio optimization or underwriting cycle management. ▪ Addressing only one side of the cycle – soft or hard but not both. A few candidates mixed up soft and hard market ▪ Many candidates interpreted "asset" in the asset management strategy as meaning "intellectual property", discussing staff retention and expense controls. This was not the intent of the question. Although maintaining investment in intellectual property is good cycle management, it would not help stabilize earnings. ▪ Recommending an asset duration management strategy. ▪ Matching duration/need for an asset-liability to ensure cash flow. Focused mainly on liabilities – misunderstanding that this has to be done no matter how you manage the UW cycle ▪ Some just repeated the strategy to manage UW cycle i.e. increase market share when hard and decrease when soft ▪ Reinsurance as an asset management strategy. Reinsurance is an important capital management tool but reinsurance reduces volatility regardless of UW cycle.
Part b
○ Candidates generally performed better on this part. ○ Any well-explained risk that increases as a result of the strategy outlined in part (a) was given full credit (regardless of whether the answer to part (a) was correct). ○ Common errors: ▪ Suggesting that interest rate risk increases as a result of using an asset-liability duration matching strategy. This strategy reduces interest rate risk. ▪ Describing a risk that is not really a risk. E.g. "risk of increasing expense ratio from maintaining staff while shrinking market share." This is not a risk because it is a certainty of the selected strategy. ▪ Providing a risk that is not affected by the strategy, e.g. strategic risk – there is always

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

strategic risk present in business decision making.

QUESTION 23	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: C2: Describe the rationale for, methods for, and effect of managing insurance and financial risks.
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> Default is a very unlikely outcome in the far tail of the distribution of outcomes. The ERM model is probably not very accurate at this point in the distribution so using default avoidance as a reference point may not yield accurate results from the ERM model. Default avoidance mainly protects policyholder. However, other stakeholders (e.g., Shareholders) may care about large partial decreases in capital. To protect all stakeholders, need to choose more likely reference point than default.	
<i>Sample Answer 2</i> This requires selecting a capital level deep in the tail of the loss distribution, which is exactly where the loss distribution is least reliable. Default avoidance mainly protects policyholders. Shareholders can be hurt at losses lower than default level. Thus a lower level than default level may be more meaningful for the firm.	
<i>Sample Answer 3</i> Shareholders are impacted by a loss in value before the company is close to default. Capital requirements should consider protecting shareholders. Default usually happens far out into the tail of a loss distribution where the results may not be as credible. Capital requirements should be based on a credible estimate.	
Part b: 0.5 point	
<i>Sample Answer 1</i> Sufficient capital to continue servicing renewals. Sufficient capital to withstand and thrive after a catastrophe.	
<i>Sample Answer 2</i> Rating agency requirement- what level of capital is required to maintain rating Point at which capital could support renewal book	
<i>Sample Answer 3</i> Setting capital at a level that maximizes franchise value Setting capital at a level to service renewal book	